Chapter 16 exercises • Virtual machines

OS16-E01 • Modes and virtual interrupts

Guest kernel runs non-root ring 0; VMM uses root context. Guest requests disabling its maskable interrupts.

1. Should host interrupts remain disabled forever? What if a guest application requests this?
2. What gap remains when a sensitive instruction does not trap and differs from native guest semantics?

OS16-E02 • Addresses and permission intersection

Pages4096 B, GVA0x2ABC, guestPTE[2]=(GFN7, rw), EPT[7]=(MFN0x12, r).

1. Find GPA, HPA, and read/write decisions.
2. Guest changes PTE[2] to GFN8 but shadow remains VPN2→MFN12. What fails?

OS16-E03 • General nested-walk cost

Guest has3 levels, EPT4; no TLB/walk-cache hits or large pages.

1. Count table references and references including data.
2. Why not3+4=7?

OS16-E04 • Four-slot PV ring

Use ordered single-producer/consumer model, N=4, req_prod7, req_cons6, rsp_prod5, rsp_cons4.

1. Find queued, outstanding, response, used, free, and next request slot.
2. After another submission, can one more submit? What changes after consuming one response?
3. Specify descriptor publication and slot-reuse synchronization.

OS16-E05 • Finite two-layer scheduling trace

One pCPU, A/B each one vCPU, alternating10 ms; A runs0–10, B10–20, and so on. Each needs20 ms CPU.

1. When does each finish? Does average share determine finite completion time?
2. Can high guest-thread priority unconditionally remove these waits?

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Chapter 16 exercises • Virtual machines (continued)

OS16-E06 • Balloon conservation

Host 8192 MiB, VMM 1024, A/B each demand4096 resident backing, not yet fully allocated. Nominal sizes stay unchanged.

1. How much reduction is required? Account for 512 ballooned in each guest.
2. Can workloads still use every ballooned page without cost?

OS16-E07 • Sharing and COW

A:g3→h9, B:g4→h11, both contain identical bytes X.

1. Is a hash match enough to free h11? Give B's first-write ordering.
2. What if B retains an old writable TLB entry after PTE becomes read-only?

OS16-E08 • Migration convergence and pause

M512 MiB, copy128 MiB/s, distinct dirtying32 MiB/s. Perform three rounds then freeze, ignoring control cost.

1. List rounds, residual, total bytes, and pause.
2. Does dirtying128 MiB/s shrink the model?
3. What if target starts, acknowledgment is lost, and source resumes?

OS16-E09 • Snapshots and external effects

Base={0:A,1:B}, delta={1:C}. Sent mail is outside VM state.

1. Evaluate active reads and rollback to base. What happens to mail?
2. Is copying an active disk file necessarily an application-consistent backup?

OS16-E10 • Abstractions and bytecode

Requirements are testing another kernel, running an old cross-ISA binary, running Java class, and isolating two same-kernel web services.

1. Choose mechanisms and identify shared layers.
2. Trace iconst_2, iconst_3, iadd, istore_1. Explain iadd on an empty stack.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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